Setup-charged signed points for actual S3 lifting

Quadratic-twist study, PR2381

25 October 2026; signed-cache-20261025

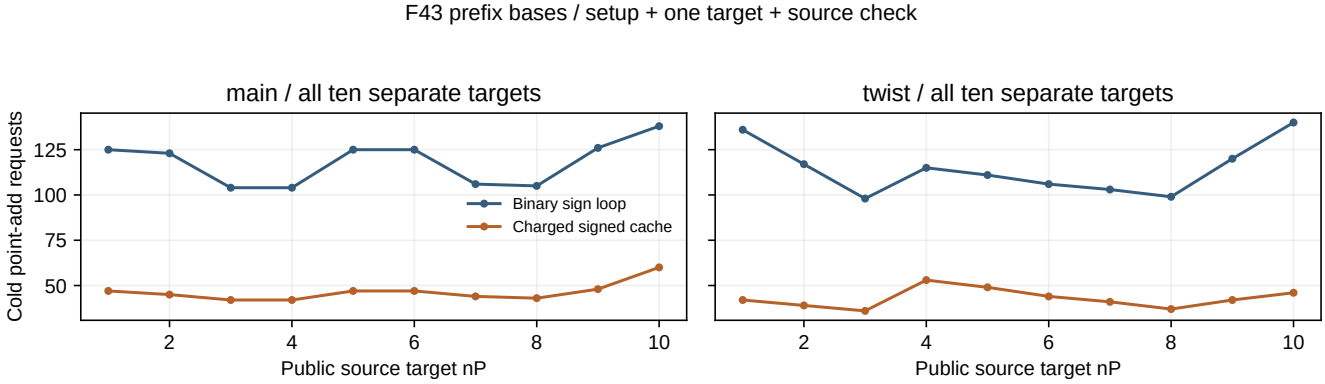
Executed comparison. C++ and independent Rust execute 80 paired setups and 800 separate nonidentity source recoveries, 40/400 per arm: ten frozen F43 size-two policies, four direct/transported roles and ten targets. All 800 scalars verify. Roots, ordered sign lifts, rows/ranks, full solver certificates, promotions, operational/expanded degrees, failed shifts, base logs and source answers stay equal. An additional batch-RREF/witness audit checks all 100 distinct solver certificates. Reference records match the frozen checked-root route exactly after dropping the new zero counters.

Cached signs, with setup included

The cache builds four signed base points, charges two negations, four stored entries, four canonical/curve checks, four bindings and two inverse-validation additions once per setup. Every tested lift reads two points and computes its actual sum. All signs are still tested, even after a successful lift in a query.

Prefix policy, separate $Q = P$	Binary main	Cached main	Binary twist	Cached twist
Setup point-add requests	79	33	73	27
Online point-add requests	44	12	61	13
Cold point-add requests	125	47	136	42
Cold negation requests	20	2	24	2
Cold scalar-loop iterations	54	14	56	8
Cold cache reads	0	40	0	48
Tested signed lifts	20	20	24	24
Represented matrix slots	540	540	36	36

All 400 candidate cold records have fewer point-add requests than their own reference, including cache validation. With L tested lifts, the exact delta is $-4L+2$ in setup/cold and $-4L$ online. Negations change by $-L+2$ (setup/cold), scalar iterations by $-2L$, and cache reads by $+2L$. Other old components are unchanged. Ratios $125/47$ and $136/42$ describe point-add requests, *not speedups*.



signed-cache-20261025; cache validation included; reads/curve checks separately counted, not priced

Reference quality matters. The old binary sign loop includes unused final doublings and identity additions. Compilers may optimize unused arithmetic. These36 unpriced components are algorithm requests, not retired instructions or runtime. Cache storage/reads, curve-check coordinate work, field tables, allocation, sorting/traversal, hashing and audit remain unpriced. Direct uncached signs are the next stronger reference. No calibrated total, GAE, S, rho, intrinsic regularity or fastest-external claim is admitted.

Validation implies exactly the old lifting rule

In an additive commutative group, validation requires

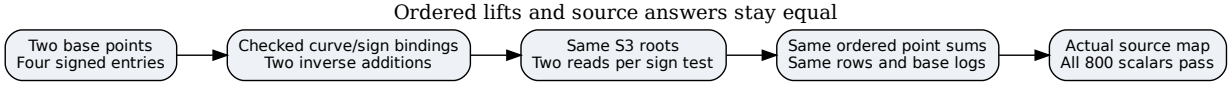
$$C_i(+) = B_i, \qquad C_i(-) + B_i = O.$$

Lean derives $C_i(-) = -B_i$, then proves equality of cached and scalar signed sums and of *ordered filtered candidate lists*, preserving order and duplicates. Arbitrary indices may repeat. A cached relation to the shifted target yields source recovery through an injective additive map; reduction modulo*r* explicitly uses the source annihilator $rP = O$.

The actual F43 twist cache is $(15,9),(17,6)$ and negatives $(15,34),(17,37)$. Mathlib nonsingular points, inverse checks and base logs $B_0 = 5U, B_1 = 7U$ are kernel checked. The concrete recovery theorem requires the actual target relation, but assumes neither cache correctness nor the base logs.

Claim	Declaration in `SignedCache`
Sign and validated cache	`signed_eq_zsmul`, `validated_cache_value`
Pair sums and ordered lifts	`cached_sum_eq`, `ordered_lifts_eq`
Shifted source recovery	`cached_relation_recovers_mod`
Actual F43 checks	`F43.actual_cache_valid`, `actual_base_logs`, `actual_cached_recovery`
Coordinate inverse	`coordinate_inverse`

Nine new public theorems; unmodified scoped gate 191 public /249 audited, zero failing/extra trusted axioms. Only propext, Classical.choice and Quot.sound. Certificate `IDC1h202bb532af10c977` checks only $y + (-y) = 0$ at 32+16 points, with false identity/source rejection. Group/list/recovery claims rest on Lean, while native execution and request vectors are independently replayed.



All policies, actual map and limitations

Across 100 unmatched cached main/twist records, twist point-add requests are lower/equal/higher in 43/1/56, matrix slots 44/15/41, scalar iterations 33/10/57. Summed hypothetical cold additions are 4398/4638, versus 11750/11990 before; these sum separate one-target records, not batch throughput. Transported bases match every component except explicit coordinate maps. Later losses remain.

Models `icv1-fp6-t0-74779950` and `icv1-fp6-t0-c6a8a47c` are $E : y^2 = x^3 + x$ and $T : Y^2 = X^3 + 4X$ over F43, order-11 generators$(31,18)$ and$(24,26)$. The actual map$(x,y) \mapsto (41x, 11y)$ binds the source DLP. This is an exceptional base-field-isomorphic fixture; the new generator has no EC1 registration. Recovery receives only the source point. Earlier first query rank0/2 and operational degree3/2 remain, with equal homogeneous cutoff2.

Seven rehashed controls reject without receipt, including an on-curve point in the wrong sign slot. The exact fresh replay and 22 backend tests pass. Both proof drafts/logs remain; final proofs generalize the first compiled fixed-column draft before native execution. Profile probe lacks gawk, Conductor exits127, and no upload or paid resource occurred. Next compare direct uncached signs on the same targets. Calibration, general geometry, stationary useful yield, review, exact-head CI access and integration remain open. Full sources, proof map, commands and validation are linked in the README.